

RESEARCH PROJECT 01
FEDERICO LOLLI AI RESEARCH

Symmetric Factorized Quadratic Adaptation

A Controlled Study of Nonlinear Low-Rank Adapters for GPT-2

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Abstract

Parameter-efficient fine-tuning adapts pretrained models through small modules while leaving the backbone frozen. This study asks whether a low-rank adapter gains useful capacity by replacing the linear LORA correction with an explicit quadratic function of projected activations. The proposed SYMMETRIC adapter projects an activation to a low-dimensional space, squares its coordinates element by element, and maps them back; it has the same trainable matrix shapes and parameter count as matched LORA. An independent 24-run WikiText-2 confirmation covers ranks 1, 2, 4, and 8, three seeds, and 5,000 optimizer steps. SYMMETRIC obtains lower mean final validation and held-out test loss at every tested rank, with all twelve paired test comparisons favouring it. A six-run AG News classification study provides the counter-result: LORA achieves higher mean accuracy and macro-F1. A single-seed scaling ablation and fresh convergence trajectories qualify the rank result. Analytical and empirical studies show that LORA has an input-independent Jacobian and zero adapter-only Hessian, whereas SYMMETRIC has an activation-dependent local Jacobian and explicit second-order response. A matched rank-4 benchmark finds nearly identical latency and a 1.25 MiB increase in peak allocated VRAM. The evidence supports quadratic activation-space adaptation as a viable, task-dependent design point rather than a universally superior replacement for linear low-rank adaptation.

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1 Introduction

Full fine-tuning updates every model weight, whereas parameter-efficient fine-tuning (PEFT) freezes most pretrained parameters and learns a small correction. This reduces task-specific state and makes the correction’s functional form a meaningful experimental variable. LORA [1] is a natural baseline: it is simple, parameter efficient, and interpretable as one constant low-rank weight update. Its adapter branch is nevertheless linear in the incoming activation and cannot itself express second-order feature interactions.

This study evaluates a small change in function class. The incoming activation is projected to r coordinates, those coordinates are squared element by element, and the result is projected back. This preserves the trainable shapes of LORA while making the correction quadratic and activation dependent. A controlled comparison is necessary because rank, parameter count, placement, scaling, training duration, seed, and task can all alter the observed ordering.

1.1 Research question

Can a low-rank adapter gain useful expressive capacity by replacing the purely linear LORA correction with an explicit quadratic function of projected activations, while retaining a comparable parameter budget and practical computational cost? The question matters because adapters of identical size can implement different local transformations.

1.2 Contributions

This work provides (i) a precise symmetric factorized quadratic activation-space adapter; (ii) controlled comparisons with LORA under matched budgets; (iii) rank, scaling, convergence, placement, and downstream analyses; (iv) local-Jacobian, effective-rank, and adapter-Hessian analyses; (v) a matched cost benchmark; and (vi) conservative positioning relative to verified related methods. These are empirical and analytical contributions, not a claim of universal novelty or superiority. Sections 2–4 introduce the context, method, and protocols; Sections 5–8 present results, mechanism analysis, and cost; Sections 9–11 discuss scope and conclusions.

2 Related work and positioning

Table 1 compares mechanisms rather than incompatible benchmark scores. Only LORA is reproduced as a direct baseline. LORA learns a linear update [1]. DoRA separates weight magnitude and direction [2]; MoRA uses a square matrix and fixed transforms [3]; HiRA forms a Hadamard-product update [4]; and LoRAN transforms a low-rank weight update nonlinearly [5]. These operate in weight or parameter space rather than by squaring the current activation.

2.1 PERA

PERA expands low-rank factors polynomially before composition [6]. This study follows $x \rightarrow xU \rightarrow (xU) \odot (xU) \rightarrow P$. Both investigate higher-order structure, but the operation acts on different objects. PERA is not an experimental baseline here.

Table 1: Mathematical and architectural positioning. External methods were not experimentally reproduced.

Method	Core mechanism	Relation to this study
LoRA	Low-rank linear correction	Direct baseline
DoRA	Weight magnitude/direction decomposition	Different weight parameterization
MoRA	Square trainable matrix with fixed transforms	Weight-space high-rank strategy
HiRA	Hadamard-product high-rank update	Weight-space higher-rank adaptation
LoRAN	Nonlinear low-rank weight update	Nonlinear weight-space method
PERA	Polynomial expansion of low-rank factors	Higher-order parameter/factor-space method
QuadraNet V2	Factorized quadratic neural transformation	Related quadratic-form precedent
This study	Squared projected activation	Activation-dependent quadratic correction

2.2 QuadraNet V2

QuadraNet V2 is a verified precedent for factorized quadratic transformations [7]. For one output, the present adapter is xQ_jx^T . Across a multi-output layer, all Q_j share directions in U and differ through coefficients in P ; single-output similarity does not establish full architectural equivalence.

2.3 Position of the proposed method

The defining feature is the element-wise square after the activation projection and before the output projection. This yields an input-dependent local transformation at matched LoRA size. Figure 9 later summarizes where higher-order operations enter the compared methods.

3 Method

3.1 Frozen projection and parallel correction

For row activation $x \in \mathbb{R}^{1 \times d}$, frozen $W \in \mathbb{R}^{d \times d_{out}}$, and frozen bias b ,

$$y_{base} = xW + b. \quad (1)$$

The adapter is a parallel trainable branch. Matched LoRA and SYMMETRIC corrections are

$$\Delta y_{LoRA} = \frac{\alpha}{r}(xA)B, \quad (2)$$

$$\Delta y_{Sym} = \frac{\alpha}{r}((xU) \odot (xU))P, \quad (3)$$

where $A, U \in \mathbb{R}^{d \times r}$ and $B, P \in \mathbb{R}^{r \times d_{out}}$. Both contain $r(d + d_{out})$ parameters, or $2dr$ for square GPT-2 c_proj. LoRA applies two linear projections; SYMMETRIC forms $z = xU$, computes $z \odot z$, then applies P .

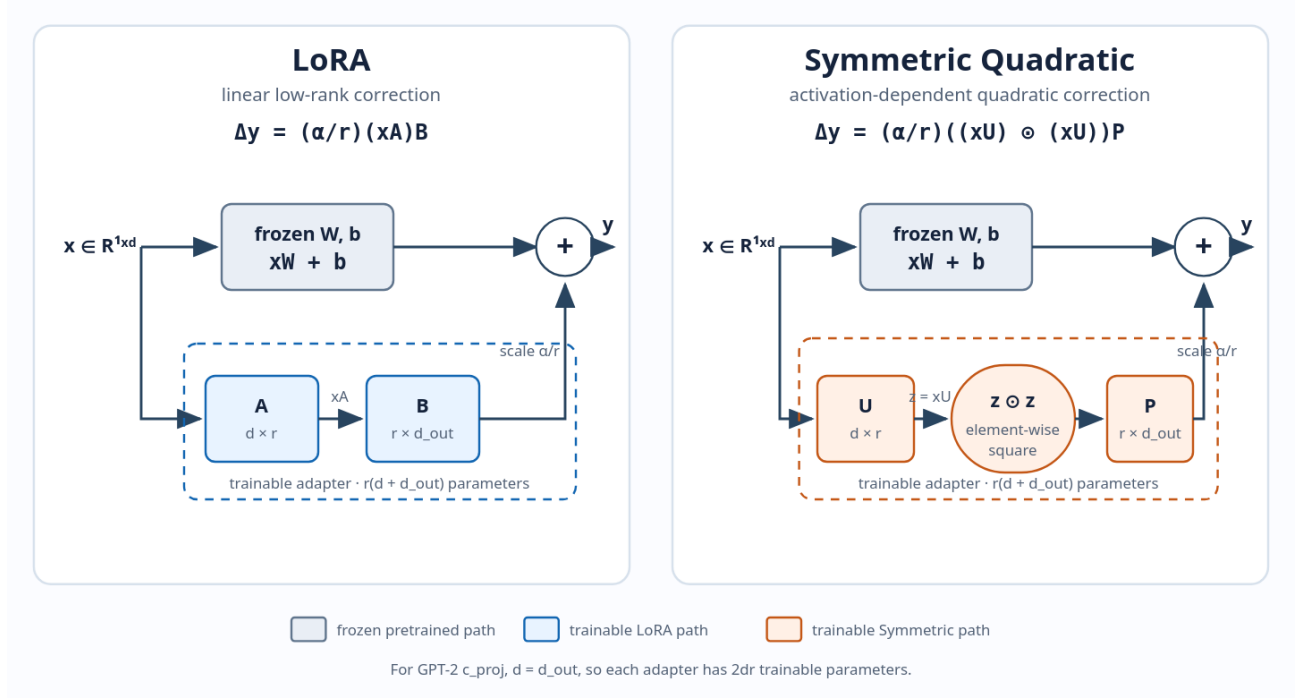


Figure 1: Architecture of matched LoRA and SYMMETRIC adapters. Both add a low-rank trainable branch beside a frozen projection. LoRA is linear; SYMMETRIC inserts an element-wise square before the output projection.

3.2 Quadratic structure

For output j ,

$$\Delta y_j = \frac{\alpha}{r} \sum_k P_{kj} (x u_k)^2 = x Q_j x^T, \quad (4)$$

$$Q_j = \frac{\alpha}{r} U \text{diag}(P_{:,j}) U^T. \quad (5)$$

Q_j is symmetric and $\text{rank}(Q_j) \leq r$. Directions in U are shared across outputs; signed coefficients in P mean Q_j need not be positive semidefinite. The identity $(ax_1 + bx_2)^2 = a^2x_1^2 + 2abx_1x_2 + b^2x_2^2$ shows that squared projections create cross terms. Output factors are zero initialized. The main study uses $\alpha = 4$, giving $\alpha/r = 4, 2, 1, 0.5$ at ranks 1, 2, 4, 8 and motivating Section 6.

3.3 GPT-2 insertion point

The controlled adapter receives the concatenated attention output entering zero-based block 0 `attn.c_proj`. Frozen projection and adapter outputs are summed before attention residual dropout and residual addition (Figure 2).

Final-study adapter location in a zero-based GPT-2 block

Block 2 is the third block. Frozen modules are gray; trainable adapter is blue.

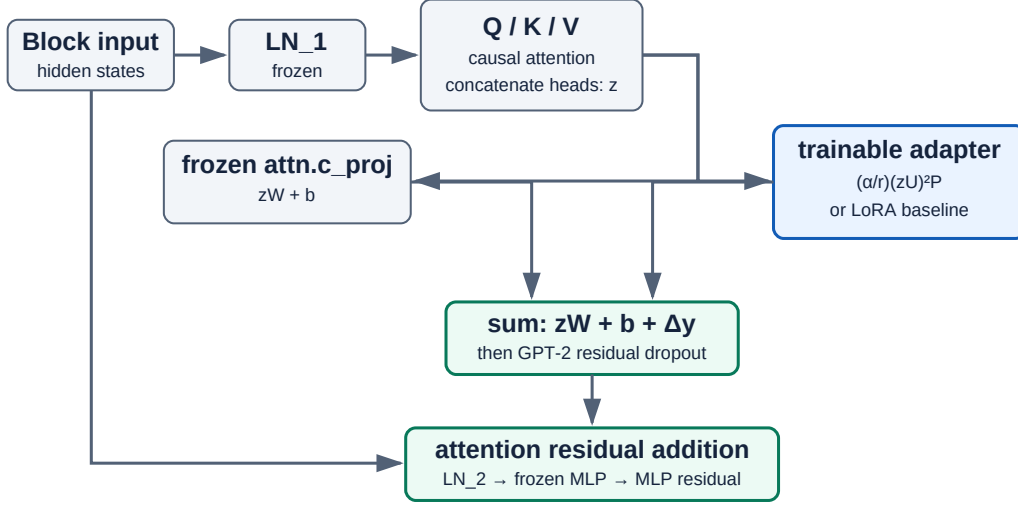


Figure 2: Controlled GPT-2 placement. The trainable adapter and frozen block-0 `attn.c_proj` share an input; their outputs are summed before residual dropout and the attention residual addition. Block numbering is zero based.

4 Experimental design

Historical studies motivated controlled questions but are not pooled with them. Table 2 records each campaign’s role.

Table 2: Completed experimental campaigns.

Campaign	Design	Role
Adapter families	Linear, quadratic, interaction, symmetric, mixed variants	Exploration
Depth/placement	Attention/MLP screens and 12-block scan	Placement
Fixed-budget multi-layer	6,144/12,288 parameters across layers	Depth/local rank
AG News	Two methods, three seeds, 500 steps	Classification
Rank screening	Two methods, four ranks, three seeds, 1,000 steps	Screening
Rank confirmation	24 fresh runs, 5,000 steps	Central LM evidence
Scaling ablation	$\alpha = 4$ versus $\alpha/r = 1$, seed 42	Scale check
Mechanism/cost	Stored checkpoints; matched rank-4 benchmark	Structure/cost

4.1 Historical exploration

Early adapter-family studies, 500/1,000/2,000-step durations, attention and MLP screens, a 12-block attention scan, and fixed-budget multilayer designs established trainability and motivated later controls. Their nonuniform protocols remain separate; Appendix B summarizes verified values.

4.2 Controlled WikiText-2 protocol

GPT-2 Small is frozen. WikiText-2 is packed into consecutive 128-token blocks for causal cross-entropy. Runs use block 0 `attn.c_proj`, batch 1, accumulation 4, AdamW at 3×10^{-4} , clipping 1.0, no scheduler, $\alpha = 4$, ranks 1/2/4/8, seeds 42/123/456, and 5,000 steps. The 1,000-step screen is independent. Best recorded validation, exact final validation, and final-checkpoint held-out test are distinct; test data select nothing.

4.3 Controlled AG News protocol

The same frozen backbone and placement are used with a classification head, 4,096 training examples, a fixed 1,000-example stratified validation subset, the original test split, rank 4, three seeds, and 500 steps. Each method has 9,216 trainable parameters including the head.

4.4 Scaling ablation

A seed-42, 1,000-step ablation compares fixed $\alpha = 4$ with constant $\alpha/r = 1$ ($\alpha = r$). It is exploratory and separate from the main confirmation.

4.5 Mechanism and cost protocols

Final checkpoints are reused for spectra, output covariance, local Jacobians, and adapter-only Hessians on fixed held-out activations. Analytical derivatives are checked against double-precision autograd. The cost benchmark uses rank 4, batch 1, sequence length 128, 10 warm-up and 30 measured iterations. No full-block Hessian was completed.

5 Controlled results

5.1 WikiText-2 confirmation

Table 3 reports final-checkpoint means and sample SD. Lower is better. Paired test differences

Table 3: Final WikiText-2 loss over three seeds.

Rank	LoRA val	SYMMETRIC val	LoRA test	SYMMETRIC test
1	3.5734 ± 0.0051	3.5314 ± 0.0207	3.9256 ± 0.0009	3.9131 ± 0.0042
2	3.4773 ± 0.0057	3.4340 ± 0.0249	3.8492 ± 0.0025	3.8305 ± 0.0144
4	3.3483 ± 0.0093	3.3101 ± 0.0011	3.7656 ± 0.0054	3.7429 ± 0.0024
8	3.3054 ± 0.0117	3.2446 ± 0.0096	3.7319 ± 0.0035	3.6938 ± 0.0044

(SYMMETRIC minus LoRA) are -0.0125 ± 0.0051 , -0.0187 ± 0.0119 , -0.0227 ± 0.0035 , and -0.0381 ± 0.0072 at ranks 1,2,4,8; all 12 pairs favour SYMMETRIC. Figure 3 shows individual seeds and SD. This result is specific to the tested task, placement, schedule, and scale.

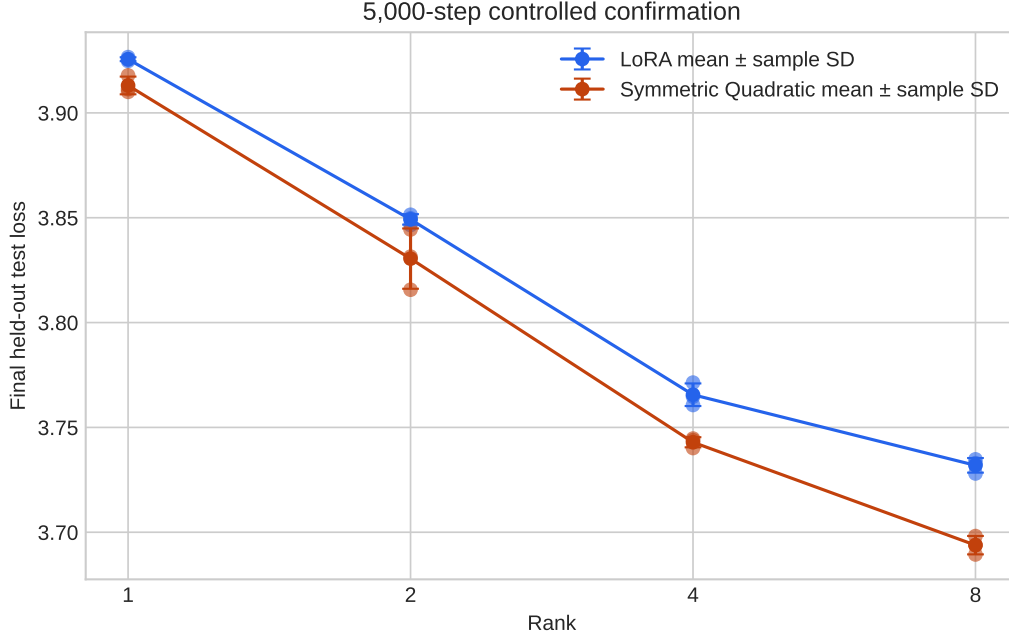


Figure 3: Held-out WikiText-2 test loss after 5,000 steps at ranks 1,2,4,8. Lower is better. Points are seeds; markers/error bars are mean/sample SD. SYMMETRIC is lower on average at every rank in this protocol.

Table 4: AG News held-out results, mean $\pm$ sample SD.

Method	Accuracy	Macro-F1	Test loss
LoRA	0.8201 ± 0.0051	0.8156 ± 0.0050	0.5445 ± 0.0542
SYMMETRIC	0.8021 ± 0.0240	0.7929 ± 0.0315	0.6847 ± 0.1919

5.2 AG News

LoRA is better on average for all three measures. Figure 4 exposes the larger SYMMETRIC seed variability, demonstrating that the WikiText-2 ordering does not transfer automatically.

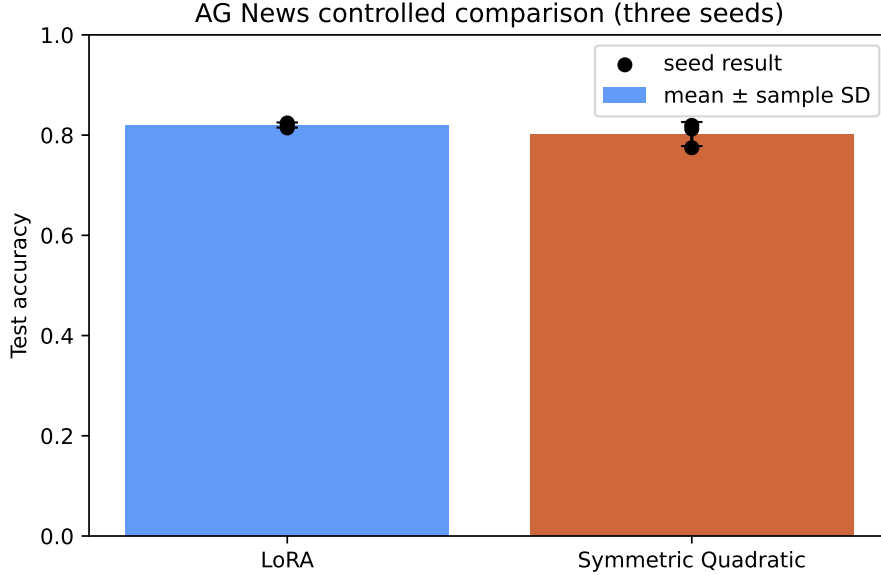


Figure 4: AG News held-out accuracy. Bars/error bars are mean/sample SD; black points show all seeds. LoRA is higher on average.

6 Scaling and convergence

6.1 Constant-scale ablation

Table 5: Seed-42 validation loss after 1,000 steps; $\Delta = \text{SYMMETRIC} - \text{LoRA}$.

Rank	$\alpha = 4$			$\alpha/r = 1$		
	LoRA	SYMMETRIC	Δ	LoRA	SYMMETRIC	Δ
1	3.6171	3.6614	+0.0443	3.6301	3.6711	+0.0410
2	3.5861	3.5561	-.0300	3.5984	3.5693	-.0292
4	3.5530	3.4780	-.0750	3.5530	3.4780	-.0750
8	3.5492	3.4315	-.1177	3.5056	3.4081	-.0975

The sign pattern persists for seed 42 while magnitude changes (Figure 5); one seed prevents broader inference.

6.2 Fresh confirmation trajectories

Table 6: Mean paired validation difference SYMMETRIC – LoRA.

Rank	1,000	2,000	3,000	4,000	5,000
1	+.0280	+.0096	-.0043	-.0265	-.0419
2	-.0367	-.0432	-.0319	-.0311	-.0434
4	-.0707	-.0631	-.0803	-.0635	-.0382
8	-.1247	-.1156	-.0947	-.0781	-.0608

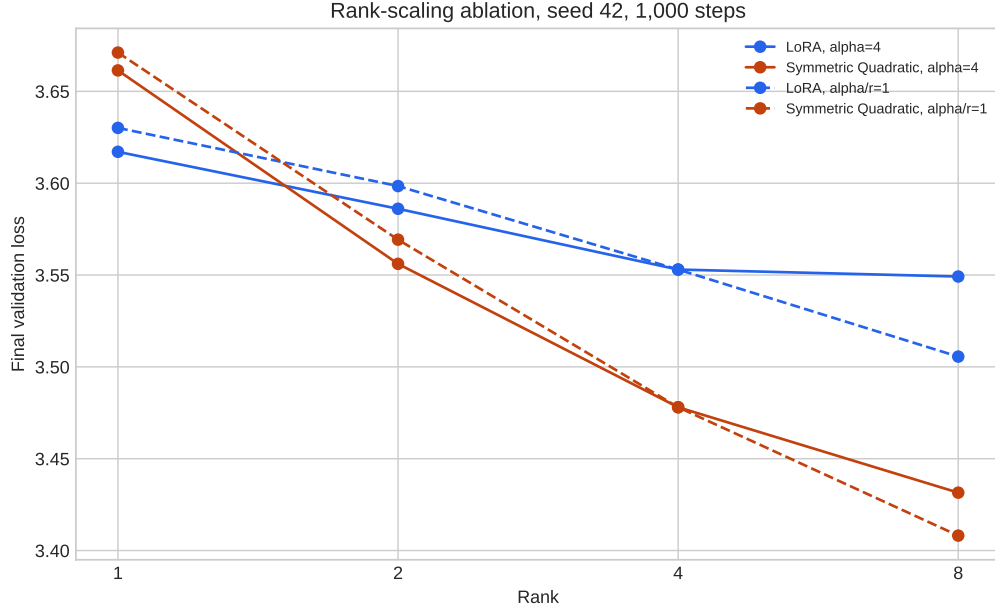


Figure 5: Single-seed scaling ablation. Fixed α changes effective scale with rank; the comparison fixes $\alpha/r = 1$.

Negative favours SYMMETRIC. Rank 1 crosses between 2,000 and 3,000 recorded steps; ranks 2–8 favour SYMMETRIC at every listed point. The independent screen is not joined to these curves. Figure 6 shows rank 1 and SD; no asymptotic claim is made.

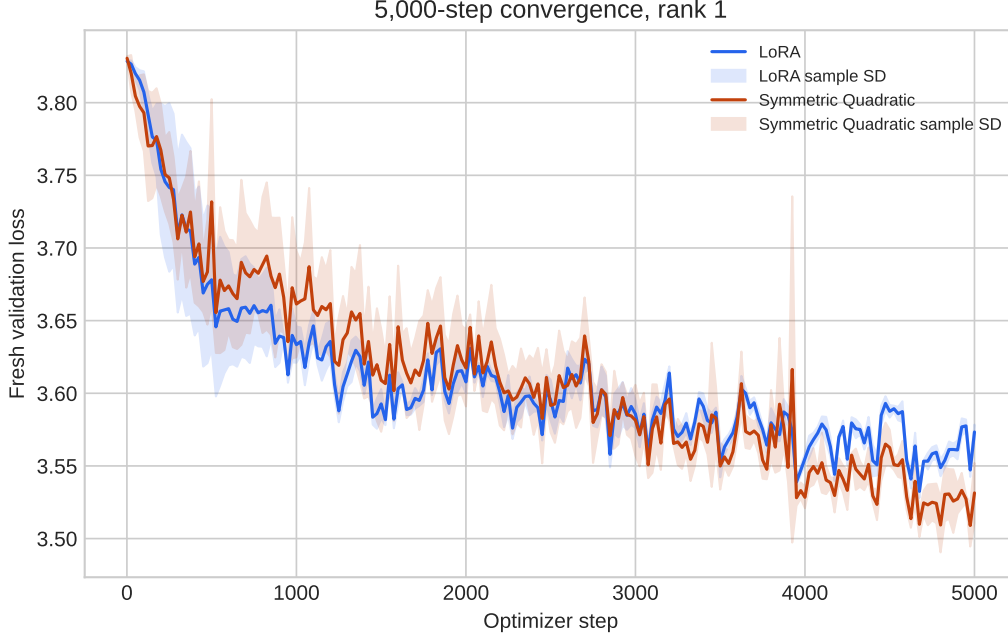


Figure 6: Fresh rank-1 validation trajectories in the 5,000-step confirmation. Curves are means; bands are sample SD.

7 Mechanism analysis

These analyses characterize structure but do not establish why performance differs.

7.1 Local Jacobians

$$J_{\text{LoRA}} = \frac{\alpha}{r} AB, \quad (6)$$

$$J_{\text{Sym}}(x) = 2\frac{\alpha}{r} U \text{diag}(xU)P. \quad (7)$$

LoRA is input independent and represents one fixed local update. SYMMETRIC changes with x and cannot generally be reduced to one constant ΔW . Double-precision tests match these expressions to autograd.

7.2 Effective rank

Entropy effective rank is $\exp(-\sum_i p_i \log p_i)$ for normalized singular values p_i . Table 7 reports distinct objects. The SYMMETRIC local-Jacobian rank is not a fixed weight-update rank directly equivalent to LoRA (Figure 7).

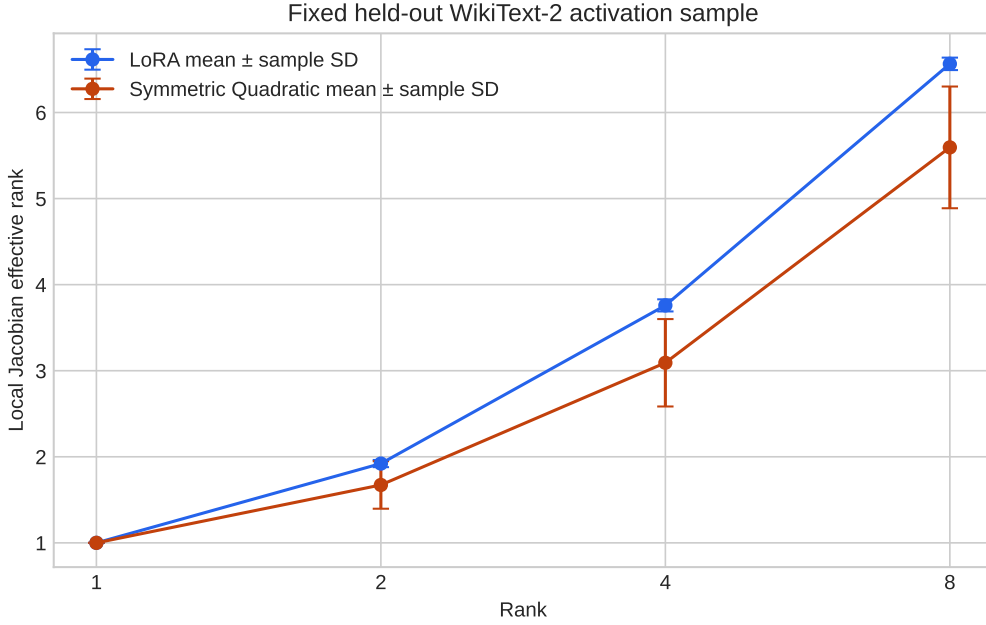
7.3 Second-order structure

For fixed linear output reduction c ,

$$H = 2\frac{\alpha}{r} U \text{diag}(Pc)U^T. \quad (8)$$

Table 7: Mean entropy effective ranks.

Rank	LoRA update	SYMMETRIC Jacobian	U	P	Output cov.
1	1.00	1.00	1.00	1.00	1.00
2	1.92	1.67	2.00	1.98	1.92
4	3.76	3.09	3.99	3.89	3.00
8	6.57	5.59	7.96	7.28	3.59

**Figure 7:** Effective rank of the LoRA update and SYMMETRIC local Jacobian. These measure related local complexity but are not identical weight-space objects.

LoRA has zero adapter-only Hessian; SYMMETRIC is nonzero. Mean SYMMETRIC Frobenius norms are 155.15, 92.13, 52.68, 17.61 at ranks 1,2,4,8. The decrease combines learned factors and scale and is not evidence of weaker useful interactions. For fixed c , the Hessian is input independent, so repeated tokens are not independent observations. Autograd verifies Equation 8; a full-block Hessian was not evaluated.

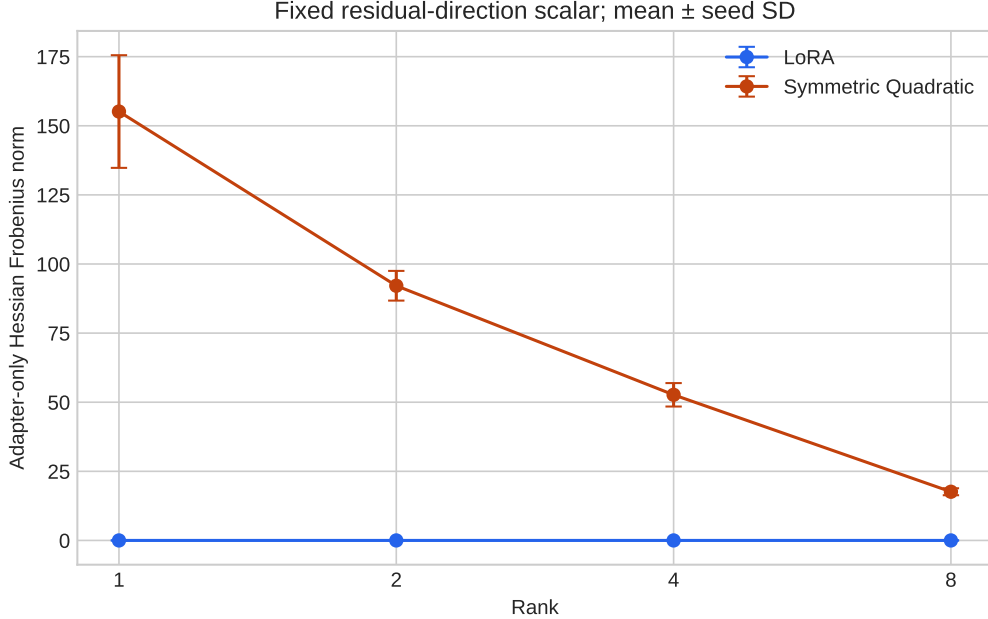


Figure 8: Adapter-only second-order response. LoRA is exactly zero; SYMMETRIC has an explicit nonzero response. This does not prove a causal performance mechanism.

8 Computational cost

Both rank-4 adapters have 6,144 parameters. Table 8 shows nearly identical measured cost. Relative

Table 8: Matched rank-4 benchmark: batch 1, length 128, 10 warm-up and 30 measured iterations.

Method	Forward (ms)	Backward (ms)	VRAM (MiB)	Tokens/s
LoRA	12.157 ± 0.243	14.805 ± 0.236	429.05	10,529
SYMMETRIC	12.184 ± 0.236	14.766 ± 0.188	430.31	10,505

differences are approximately +0.22% forward latency, −0.27% backward latency, +1.25 MiB VRAM, and −0.22% throughput. The result applies only to this implementation and hardware configuration.

9 Discussion

The controlled WikiText-2 result is positive and configuration specific: SYMMETRIC is lower at every rank and in every paired held-out comparison. AG News favours LoRA on average, showing task dependence. Rank and optimization duration matter: rank 1 changes ordering during training, while higher-rank gaps narrow over the recorded interval. Constant scale alters magnitude but does not change the tested seed’s qualitative pattern; the design does not fully isolate rank, scale, and parameter count.

The derivative study guarantees a structural distinction and measures learned spectra, but does not prove that second-order response causes lower language-model loss. The cost study shows that this structure can be implemented at almost the same measured practical cost. Figure 9 summarizes the architectural context without implying equivalence.

Where the adaptation is applied

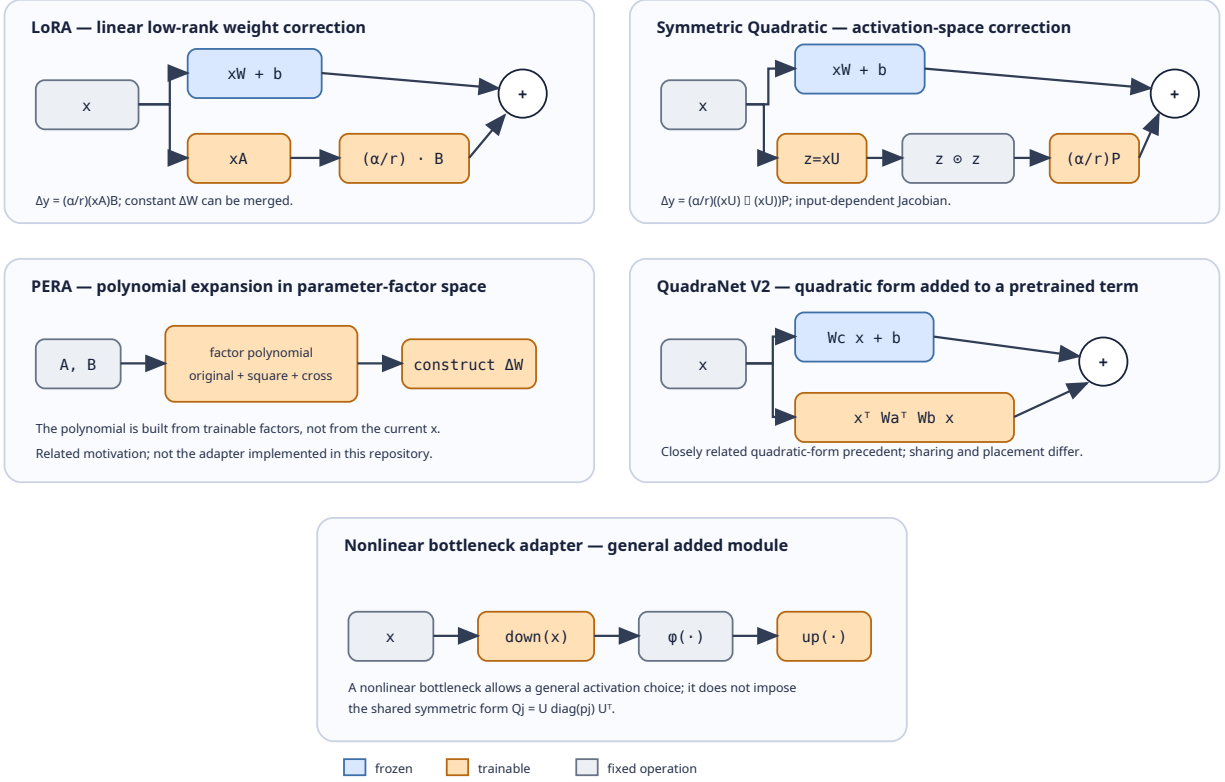


Figure 9: Architectural comparison of LORA, SYMMETRIC, PERA, QuadraNet V2, and a nonlinear bottleneck adapter. The location of a higher-order operation differs; visual similarity does not imply mathematical equivalence.

10 Limitations

Evidence is limited to GPT-2 Small, two datasets, restricted placements, three central seeds, one scaling-ablation seed, and no exhaustive hyperparameter sweep. Related methods except LORA were not reproduced. There is no full-block Hessian, and cost was measured for one rank, device, batch size, and sequence length. The literature review does not establish universal novelty over every quadratic architecture.

11 Conclusion

SYMMETRIC is a viable quadratic activation-space adapter. Under controlled 5,000-step WikiText-2 conditions, it achieves lower mean final validation and held-out test loss than matched LORA at ranks 1,2,4,8. AG News favours LORA on average, so the evidence is task dependent. Rank, scaling, and duration qualify the result. Jacobian and Hessian analyses establish functional differences, while the matched benchmark finds small overhead. Adapter functional form therefore matters alongside parameter count, but broader generalization requires additional models, tasks, placements, and replications.

A Per-seed results and provenance

Per-seed WikiText-2 values are available in `results/tables/rank-confirmation-5000/rank_5000_per_seed.csv`; AG News runs are in `results/tables/downstream-ag-news/downstream_controlled_results.csv`. Aggregate tables in this paper are reconstructed from those files without cherry-picking. Figure provenance is recorded in `results/manifests/FIGURE_MANIFEST.md`.

B Historical evidence

Historical best-recorded validation summaries include, at 500 steps: LORA 3.5732, element-wise quadratic 3.6351, signed quadratic 3.6257, interaction 3.5581, SYMMETRIC 3.5526, and Linear+Quadratic 3.5302. At one rank-4 insertion point, 1,000-step summaries were 3.5491, 3.4770, 3.4928 for LORA, SYMMETRIC, Linear+Quadratic; at 2,000 steps they were 3.4598, 3.3784, 3.4360. These exploratory results motivated controls but are not merged with confirmation trajectories.

C Reproducibility notes

Canonical tables are under `results/tables/`; commands and recorded software/hardware details are in `research/reproducibility/REPRODUCIBILITY.md`. Analytical Jacobian, Hessian, spectral, aggregation, and public-data tests are part of the repository suite. Raw private checkpoints are not embedded in the public manuscript.

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